

DBCLS BioHackathon 2026 report: Extending TogoMCP beyond RDF Portal while making its schema guides check their own answers

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Abstract

TogoMCP is a Model Context Protocol (MCP) server that lets large language model (LLM) agents query life-science knowledge graphs in SPARQL, guided by per-database schema documents called MIE (Metadata Interoperability Exchange) files. During the DBCLS BioHackathon 2026 (BH26), the TogoMCP group set out to extend the server, refine the MIE files, add databases, learn what makes a good SPARQL example, and turn use cases into reusable agent skills. In the first six days of the event we published eleven releases (v2.12.2 to v2.20.0) and grew the catalogue from 37 to 45 databases: Fanta.bio, WikiPathways, IDSM, PubCaseFinder, LIPID MAPS, SwissLipids and MarpolBase, plus BH26 Microbes, an experimental dataset of KEGG Orthology assignments for 57.6 million prokaryotic proteins built for the BioHackathon. Five of them are served from endpoints outside RDF Portal, and a sixth from a QLever engine inside it rather than Virtuoso, and they broke assumptions the rest of the corpus had taught, such as named-graph pinning, federation with SERVICE, literal typing, and even that a result set is complete. For the first time, MIE files were written by people other than the server's maintainer, one of them by the maintainer of the database it describes, and a second group member built TogoCX, a companion MCP server that returns database edges with instructions on how their claims may be stated. We made every worked example in the MIE corpus (now 420) assert its recorded result against the live endpoint in continuous integration. The first full comparison found drift that execution-only checks had missed, most seriously a NANDO release that made an example join silently miss about 88% of mapped diseases. Onboarding the new databases uncovered a series of quantified "silent wrong answers", queries that return plausible results rather than errors, including a category count inflated 3.1-fold by an upstream data defect and a gene enumeration that returns 10,000 of 18,080 rows without an error or a warning. We also added two PubCaseFinder tools for phenotype-driven rare-disease diagnosis support, began serving analysis workflows (agent skills) from the server, fixed failure modes found in production call logs, made TogoID errors suggest working conversion routes, and prepared a LOTUS natural-products graph for hosting on RDF Portal. We argue that for LLM-facing schema documentation, "the example still runs" is not evidence that it is still correct.

Keywords: TogoMCP; Model Context Protocol; SPARQL; RDF; knowledge graphs; LLM agents; schema documentation

Introduction

Life-science data are increasingly published as Resource Description Framework (RDF) graphs behind SPARQL endpoints, but writing correct SPARQL against an unfamiliar schema remains a barrier for most researchers. TogoMCP addresses this by exposing SPARQL endpoints, REST search services and ID conversion to LLM agents through the Model Context Protocol ([MCP](#)), and by giving the agent a compact, database-specific schema guide, the MIE file, before it writes a query (Kinjo et al., [2026](#)). At the DBCLS BioHackathon 2025 we redesigned the MIE format around executable examples after an ablation study showed that the query-construction material carries almost all of the benefit (Kinjo & Yamamoto, [2026](#)). The public server runs at togomcp.rdfportal.org.

At the start of BH26 (v2.12.1, 3 September 2026), TogoMCP covered 37 databases. Thirty-five of them were served from RDF Portal endpoints operated by DBCLS; the other two were TogoVar and GlyCosmos. The TogoMCP hacking group registered five objectives for the week:

1. extending and enhancing TogoMCP;
2. examining and refining MIE files;
3. finding interesting use cases and workflows, and turning them into skills;
4. learning how to make good SPARQL examples;
5. adding new databases to TogoMCP.

This report describes what we did toward each objective between 13 and 18 September 2026, and what we learned about keeping LLM-facing schema documentation correct as the set of databases, endpoints and authors grows.

New databases

Table 1 summarises the databases added during the week. Each was onboarded the same way: a row in the endpoint registry, an MIE file with live-verified examples and documented traps (“gotchas”), and a review pull request. Falsifiable claims carry a `check`: block that can be re-run against the endpoint.

Table 1: Databases added during BH26; the version in the first column is the TogoMCP release that added the database. CRE, cis-regulatory element; HCO, Human Chromosome Ontology; HPO, Human Phenotype Ontology; KO, KEGG Orthology; SIB, Swiss Institute of Bioinformatics. Counts are as measured when each MIE was verified.

Database (release)	Content	Endpoint	Notable for agents
Fanta.bio (2.13.0)	821,722 human and mouse CREs with linked genes and ChIP-Atlas peak overlaps	RDF Portal primary	chromosome IRIs spelled differently from HCO
WikiPathways (2.14.0)	2,087 pathways across 39 organisms	WikiPathways' own Virtuoso	revision-pinned pathway IRIs; SERVICE works
IDSM (2.14.0)	structure search over nine small-molecule datasets	ELIXIR CZ, PostgreSQL-based	union default graph (×30 rows unpinned)
PubCaseFinder (2.15.0)	18,375 OMIM/Orphanet diseases, 379,263 HPO annotations	RDF Portal primary	only Japanese HPO and disease labels

Database (release)	Content	Endpoint	Notable for agents
LIPID MAPS (2.15.0)	52,175 classified lipids, shorthand such as PC 34:1	lipidmaps.org, behind a WAF	no named graphs; inverted
SwissLipids (2.16.0)	777,965 lipids with sn-position composition	SIB	subClassOf upstream ID-mapping examples return 0 rows
BH26 Microbes (2.19.0; experimental)	KofamScan KO assignments for 57.6M RefSeq proteins in 23,434 prokaryotic genomes	RDF Portal microbes (QLever)	24% of hits below threshold; no organism names
MarpolBase (2.20.0)	18,080 <i>Marchantia polymorpha</i> genes (MpTak_v7.1) with function, co-expression, orthogroups and curated literature	marchantia.info, its own Virtuoso	every result truncated at 10,000 rows

What each database adds

Fanta.bio holds promoters and enhancers from FANTOM, defined by cap analysis of gene expression (CAGE) and annotated with transcription factor binding peaks from ChIP-Atlas (Zou et al., 2024), ENCODE SCREEN candidate cis-regulatory elements and refTSS transcription start sites. Before it, TogoMCP could say where a gene is and which variants lie in it, but not which regulatory elements act on it. **WikiPathways** (Agrawal et al., 2024) adds community-curated pathways with signed, directed interactions. **IDSM** (Galgonek & Vondrášek, 2021) is the first database in the catalogue that searches by chemical structure: its Sachem extension (Kratochvíl et al., 2018, 2019) turns one SMILES string into the molecule's identifiers in PubChem, ChEMBL, ChEBI, Wikidata, DrugBank and four other datasets. Additionally, in 2024, IDSM was extended to integrate mass spectrometry databases, allowing for SPARQL queries based on mass spectral similarity (Galgonek & Vondrášek, 2024). **PubCaseFinder** (Fujiwara et al., 2018) contributes the knowledge base behind DBCLS's rare-disease diagnosis support, including text-mined disease–phenotype annotations based on the HPO (Gargano et al., 2024). **LIPID MAPS** (Conroy et al., 2024) is the only resource here that resolves the lipidomics shorthand used in result tables (34,567 structures map onto 6,988 shorthand strings), and **SwissLipids** (Aimo et al., 2015) is the only one that records which fatty acid occupies which sn-position, so “lipids with palmitate at sn-1” becomes a structured query. **MarpolBase** (Tanizawa et al., 2026) is the first plant genome in the catalogue and the first bryophyte: the MpTak_v7.1 reference for the liverwort *Marchantia polymorpha*, with structural and functional annotation for 18,080 genes, 18,232 cross-species orthogroups, a 740,830-edge co-expression network over 164 RNA-seq conditions, and 2,609 hand-curated gene–literature assertions across 510 papers. Those assertions hold what no other source in the catalogue does: each records, in a curator's own words, what a given paper says about a given gene, typed by the gene's role in that study (subject, background, comparator or tool) and by the kind of evidence behind it.

An experimental BioHackathon dataset

BH26 Microbes (bh26microbes) was built for this BioHackathon by Shuichi Kawashima and Yoko Okabeppu, and has not yet been publicly announced. It holds KofamScan (Aramaki et al., 2020) KEGG Orthology (KO) assignments, each with its HMM score, E-value, profile threshold and significance, for 57,612,257 RefSeq (O'Leary et al., 2016) proteins across 23,434

prokaryotic genome assemblies. It is the only source in the catalogue that answers “which genomes encode function X” from gene content: 2,246 genomes carry a significant hit to the nitrogenase iron protein NifH (K02588), and 1,839 carry both NifH and the MoFe protein alpha chain NifD (K02586). Because the dataset may still change shape or be withdrawn, the MIE and the Usage Guide catalogue mark it as experimental, and it is deliberately left off the public introduction page. Genomes and proteins carry no names or taxonomy on this endpoint. The MIE therefore resolves organisms by rewriting each RefSeq assembly accession (GCF) to its paired GenBank accession (GCA) and matching UniProt proteomes over SERVICE, and resolves proteins to UniProtKB by rewriting the RefSeq protein IRI into UniProt’s form. All nine examples pass their result assertions, and all eight gotcha checks pass against the live endpoint.

Endpoints and engines beyond RDF Portal’s Virtuoso

Five of the eight databases are served from their maintainers’ own endpoints, and BH26 Microbes is served by RDF Portal from a QLever (Bast & Buchhold, 2017) instance rather than Virtuoso. These endpoints exposed habits that the Virtuoso-hosted corpus had taught agents and authors:

- **Graph pinning.** TogoMCP’s guidance is to pin every query to a named graph, because co-hosted graphs re-declare shared ontology classes. LIPID MAPS has no named graphs, so a pinned query returns nothing. IDSM has a union default graph, so an unpinned query returns about 30 times as many rows. MarpolBase shows that the rule was not specific enough. Its endpoint hosts one database, yet that database’s own function and annotation graphs re-declare identifiers and labels on the same gene IRIs, so the canonical gene lookup returns 36,087 rows unpinned against 18,080 pinned ($\times 2.00$). Pinning “the database’s graphs” does not help when both offenders belong to the database: exactly one graph has to be named, and which one depends on the question.
- **Federation.** WikiPathways and IDSM can call RDF Portal and SIB with SERVICE; every outbound SERVICE from LIPID MAPS returned HTTP 502. Cross-database LIPID MAPS queries therefore have to be run from RDF Portal’s ebi endpoint, with LIPID MAPS inside the SERVICE clause. MarpolBase refuses SERVICE by configuration rather than by failure, in 0.1 s, with a Virtuoso privilege error, so no retry and no simpler remote pattern will succeed, and every cross-database hop has to be a second query.
- **Engine semantics.** A Virtuoso behaviour documented across the corpus (plain and xsd:string literals do not match each other) does not apply to IDSM’s PostgreSQL-based engine. Re-testing the two-argument REGEX() bug (alternation and brace quantifiers silently match nothing unless a third, flags argument is passed) across the whole registry showed it on 12 of the 16 endpoints rather than on “every endpoint”: IDSM, LIPID MAPS, SwissLipids and the QLever-backed BH26 Microbes all handle the two-argument form correctly, and the four exceptions are invisible from inside a result. QLever also rejects a query that uses any undeclared prefix (even rdf:), returns 0 rows without error for Virtuoso’s bif:contains text search, and matches string literals only in plain form although DATATYPE() reports xsd:string. It also caps memory per query, so a VALUES join over several KOs can abort where a UNION of constant-bound branches runs about ten times faster.
- **Deployment, not only engine.** MarpolBase runs Virtuoso, the same engine as RDF Portal, and still needed three new rules. Its results are truncated at 10,000 rows and a larger LIMIT does not lift the ceiling: enumerating the 18,080 genes returns exactly 10,000 rows with LIMIT 20000 and with no LIMIT at all, while COUNT over the same pattern correctly reports 18,080, and ordered OFFSET paging fails in two further ways. ORDER BY sorts xsd:double and xsd:integer literals as strings, so ORDER BY DESC(?fold) tops out at 999.8 when the true maximum is 48,226.3. And adding ORDER BY to an aggregate that has no GROUP BY makes Virtuoso group by the sort variable, turning one row of 18,080 into 18,080 rows of 1. The last of these is not a MarpolBase property at

all: it reproduces on RDF Portal's primary endpoint, so a trap found while onboarding a new database turned out to apply to databases the corpus had already documented.

- **Web application firewall (WAF).** The LIPID MAPS endpoint sits behind Cloudflare, which rejects legal SPARQL with HTTP 403 whenever the raw request body matches `\b(substr|concat|char)\b(`, even inside a string literal. `GROUP_CONCAT(` passes because the underscore defeats the word boundary, and `SUBSTR (` with a space is valid SPARQL that clears the rule. Separately, a client that sends Python's default User-Agent is rejected outright.

These differences also exposed a bug in our own tooling. `run_sparql` distinguishes "the entity is absent" from "the query is broken" by probing the anchor entity inside the query's graph; on an endpoint without named graphs that probe is always false, so broken queries were reported as true negatives. The probe now follows the query's own scoping.

Schema guides that check their own answers

Each MIE example records the result it returned when it was verified. Until this week, nothing compared that record with what the endpoint returns today. A weekly job flagged examples that errored or returned zero rows, but a query that kept running while its answer changed looked healthy. The verified: block was also free-form: across the 334 examples the corpus held at the start of the week it used 198 different key sets, mostly prose.

Assertable results

We reserved four keys that `scripts/check_mie_examples.py` asserts against the live endpoint (Listing 1):

- `n`, a single-cell result such as a `COUNT`, within a tolerance (default 2%);
- `row_count`, the number of rows, which must be below any `LIMIT`;
- `min_rows`, the honest assertion for a `LIMIT`-capped query;
- `has_values`, values that must each appear in the result (IRI local names match).

Every example must carry at least one of them. A malformed assertion is reported as `MALFORMED`, and a well-formed one that the live result no longer satisfies is reported as `DRIFT`; both fail pull-request and weekly jobs. We migrated the corpus by re-running every example and comparing the result with the old free-form record, rather than stamping today's figures over it. That comparison is what found the problems below.

Listing 1. One example from the LIPID MAPS MIE (abridged).

```
- id: lipid_core_record
question: What are the formula, monoisotopic mass, InChIKey and names
of cholesterol (LMST01010001)?
sparql: |
  SELECT ?label ?formula ?mass ?inchikey WHERE {
    <https://www.lipidmaps.org/rdf/LMST01010001>
      rdfs:label ?label ; chebi:formula ?formula ;
      chebi:monoisotopicmass ?mass ; chebi:inchikey ?inchikey . }
verified:
  row_count: 2
  has_values: [Cholesterol, C27H46O, HVYWMOMLDIMFJA-DPAQBDIFSA-N]
  date: '2026-09-15'
  note: 2 rows, not 1; the two rdfs:label values duplicate every other column.
```

What the comparison found

- **NANDO had been replaced upstream.** The 2026-05-22 NANDO release moved most MONDO mappings from `skos:closeMatch` to `skos:exactMatch`, so the MIE's MONDO join silently missed about 88% of mapped diseases. The same release added a paediatric chronic-disease programme, so notification numbers now repeat across two programmes. The file was revised against the live graph.
- **Figures had moved.** The Gene Ontology graph-pin warning quoted a 3.27-fold row inflation that is now 2.84-fold, because the set of co-hosted graphs re-declaring GO classes changed. Other figures were re-measured with dated notes, including taxonomy species (+4.9%) and HGNC EC-code cross-references (+3.8%).
- **Examples returned more than they claimed.** A ChEBI hierarchy example returned 18 rows for 5 amino acids (one row per path through `rdfs:subClassOf+`), and an anatomy example named parts that its query does not return.
- **Documentation one day old was already wrong.** MarpolBase changed shape twice during the week: an ontology namespace was retired on 17 September and a literature graph on 18 September, each leaving queries written against the previous day's description returning 0 rows with no error. Both are recorded in the MIE as traps rather than quietly corrected, because an agent may still be working from the older description.

Release 2.12.2, which carried the migration, also fixed a user-reported UniProt error of the same kind ([dbcls/togomcp#219](#)). An example fetched a protein's canonical sequence by naming the `-1` isoform, but across all 575,503 reviewed entries, 782 have a different canonical isoform, and in 613 of them a non-canonical `-1` exists. The lookup therefore returned the wrong sequence rather than no rows.

The checker itself had a bug that only an external endpoint could reveal. It ran every example against its own database's endpoint and ignored an example's `endpoint_name`. For a database hosted on RDF Portal the two are almost always the same (64 of the 65 examples that carry the key), so the bug stayed invisible until the LIPID MAPS cross-database example, which must run from `ebi`, failed on every run. Resolution now follows `run_sparql` (explicit URL, then `endpoint_name`, then the database default), and two guards assert that the override is still exercised.

Keeping examples away from the benchmark

The MIE specification forbids using a benchmark question's subject as an example's vehicle, since that would leak answers into the context we evaluate. The rule had been enforced by eye. `scripts/check_mie_leakage.py` now matches every question's keywords, answer heads and answer IDs against the examples of the databases that question uses, and runs in CI on any change to an MIE or a question. Its first run, over 100 questions and the 37 MIE files that then existed, found 19 matches, all generic vocabulary. Each was recorded as a reasoned waiver naming the database, question and token it covers, so that the checker reports the waiver as stale if its match ever disappears. The same run over today's 45 MIE files still reports 19 matches, all waived, and no leak.

Silent wrong answers

The most expensive failures we met did not raise errors. They returned plausible numbers. Table 2 lists those found while onboarding and auditing databases this week. Each is now a gotcha in the relevant MIE with its measured magnitude and a re-runnable check.

Table 2: Queries that return plausible but wrong results, found during BH26. FALDO, Feature Annotation Location Description Ontology.

Database	Trap	Measured effect	Remedy documented
LIPID MAPS	spurious upstream edge category/6 - > category/101 pulls Fatty Acids [FA01] into Prenol Lipids	8,558 instead of 2,739 ($\times 3.1$)	drop the edge in category queries
LIPID MAPS	lipid class nouns (e.g. "cardiolipin") are absent from lipid labels	label search finds 2 of 1,306 cardiolipins and 11 of 291 prostaglandins (recall 0.2–4%)	classify via headgroup shorthand or category tree
LIPID MAPS	three count keys had been merged into one figure	607 reported; 708 nodes, 672 labelled, 607 with children	three keys with set arithmetic
ChEBI and LIPID MAPS	chebi: vs chemof: property names	joins return 0 rows	join on InChIKey (1,498 of 1,500 agree)
WikiPathways	one compound filed under several ChEBI IDs	an undercount: 59 of 109 pathways for glucose (46% missed)	collect the full ID set first
WikiPathways	dcterms:isPartOf also targets interactions and complexes	an overcount: 304 instead of 107 pathways for MTOR ($\times 2.84$)	require a wp:Pathway on the parent
SwissLipids	fixed-depth hierarchy walk	a 4-hop walk misses 87% of Sphingolipids	use 1–5 hops for a category
PubCaseFinder	HP labels re-declared by three ontology graphs	$\times 3.15$ rows unpinned	pin the graph
PubCaseFinder	article-to-MeSH predicate minted as fabiohasSubjectTerm	correctly spelled predicate finds nothing	use the IRI as minted
Fanta.bio	chromosome IRI form differs from HCO	IRI join returns 0 rows	join on strings
NANDO	release moved MONDO mappings to	$\sim 88\%$ of mapped diseases missed	query both predicates
BH26 Microbes	skos:exactMatch rdfs:seeAlso protein-to-KO links include the 24.1% of hits KofamScan marks not significant	2,292 instead of 2,246 genomes for NifH; $\times 1.07$ KO pairs for <i>E. coli</i> K-12	require significant = true on the reified hit
BH26 Microbes	one identical hit statement per genome that shares a protein	$\times 1.033$ hits overall	count distinct proteins or genomes

Database	Trap	Measured effect	Remedy documented
BH26 Microbes	Virtuoso bif:contains sent to QLever	0 rows, no error	<code>FILTER(CONTAINS(LCASE(...)))</code> on KO definitions
TogoID and BH26 Microbes	RefSeq-to-UniProt conversion	RecA (NP_417179) maps only to A0A485JBB4, not Swiss-Prot P0A7G6	rewrite the RefSeq IRI and join UniProt over SERVICE
MarpolBase	every result truncated at 10,000 rows, and a larger LIMIT does not lift it	10,000 of 18,080 genes returned, while COUNT says 18,080	partition on the gene-ID prefix
MarpolBase	ORDER BY on an aggregate with no GROUP BY is implicitly regrouped	one row of 18,080 becomes 18,080 rows of 1	drop the ORDER BY, or write GROUP BY explicitly
MarpolBase	the database's own two graphs re-declare identifiers and labels on the same gene IRIs	36,087 instead of 18,080 ($\times 2.00$)	pin exactly one graph, not "the database's graphs"
MarpolBase	FALDO puts begin after end on the minus strand	a begin/end window finds 2 genes instead of 3; 50.1% of genes at risk	range on mpo:minPosition and mpo:maxPosition
MarpolBase	ORDER BY sorts numeric literals lexically	DESC(?fold) tops at 999.8 against a true maximum of 48,226.3	cast the sort key
MarpolBase	co-expression edges stored one-directionally	mpo:gene1 alone loses 97% of one gene's 200 partners; 690 genes return 0	UNION over both directions

The LIPID MAPS findings began with an audit of the new MIE that Kozo Nishida carried out with Claude, which was followed up in review ([dbcls/togomcp#231](#)). The label trap is structural rather than a data defect: every lipid carries a systematic name and a shorthand abbreviation, and the class noun a biologist would type appears in neither. Prostaglandin E2 (PGE2) has two labels, and neither contains "prostaglandin". Two routes that should agree (headgroup shorthand and the category tree) give the same 1,306 cardiolipins, so each checks the other.

The BH26 Microbes traps show that the pattern holds even for a dataset built during the week itself. The most natural query, following the direct protein-to-KO link, over-annotates because the link is asserted for every KofamScan hit, including those below the KO's adaptive score threshold; only the reified hit records which hits KofamScan accepted.

MarpolBase contributes a kind of trap the corpus had not recorded before. Two of its six are not about the data at all but about how much of it the endpoint will hand over, and neither is visible from inside the result: a truncated enumeration and a regrouped aggregate both return well-formed rows of the right shape, in well under a second, with no warning, and nothing in the response distinguishes 10,000 rows that are the answer from 10,000 rows that are the first half of one. The dataset's producers had also shipped query caveats of their own in a metadata graph, and two of them did not reproduce (a warning that a UNION neighbour query "did not

finish in 43 minutes”, measured here at 0.04 s, and a claim about casts inside FILTER), so the MIE records that they were tested and dropped, to stop a later author restoring them from the upstream source.

Rare-disease diagnosis support with PubCaseFinder

Most of the PubCaseFinder REST API (Fujiwara et al., 2022) is a front end over the RDF that TogoMCP already serves. We wrapped only the two functions that the RDF cannot reproduce.

- `pubcasefinder_rank_by_phenotypes` ranks OMIM diseases, Orphanet diseases or genes against a set of HPO terms, with PubCaseFinder’s information-content-weighted matching through the HPO hierarchy. For three Marfan-like phenotypes, a SPARQL “has all three” query finds 6 unordered diseases, whereas the ranking places 9 diseases at score 1.0 and grades every partial match below them.
- `pubcasefinder_get_case_reports` lists published case reports for a MONDO disease, in English (PubMed) or Japanese (J-STAGE). The index is built by text mining rather than MeSH indexing: 507 of Marfan syndrome’s 1,615 reports carry no Marfan MeSH heading, and the MeSH route over the PubMed RDF did not finish within 200 s.

Results carry disease names, MONDO IDs and genes, so the output of one tool feeds the other. Two operational details shaped the implementation. The documented ranking endpoint returns HTTP 404, so the tool uses the path the PubCaseFinder web application calls. DBCLS caps API use at 10 requests per minute, 100 per hour and 1,000 per day for the whole server, so the tools cache results and refuse with an explanatory error rather than exceed the quota.

Fixes driven by real usage

Production call logs

The server’s call log from 27 July to 15 September 2026 showed two classes of calls that always failed.

- **Agents used the upstream API’s parameter names.** 134 calls to the NCBI tools were rejected because agents sent E-utilities’ own names (`retmax`, `retstart`, `id`) instead of ours. The tools now accept both.
- **Clients cached old tool lists.** ChatGPT connectors use a frozen snapshot of the tool list, so some clients still called tools renamed months earlier; 107 such calls failed, including calls from one user on five separate days. Renamed tools are now served under their old names with a notice asking the user to refresh the connector, and retired tools return an error that points to their replacement.

We also rewrote the ChatGPT setup instructions after finding that the advice to “re-run Scan Tools” cannot work on several plans, where updating a tool list requires an administrator or re-creating the app. The handbook (English and Japanese) and tutorial used by participants were regenerated.

Togoid conversion errors that suggest routes

Togoid (Ikeda et al., 2022, 2025) converts identifiers between databases along pre-computed tables. When an agent asked for a pair with no direct table, it previously received a bare “no route” error. The error now lists up to five routes of at most two hops from Togoid’s `/route` endpoint, shortest first. `togoid_identifyId` gained an opt-in `verify=True` that asks `/lookup/id` which tables actually contain an identifier, since pattern matching alone says only that 672 is well formed for 17 datasets. Both endpoints were suggested by Togoid’s maintainers in response to our configuration request ([togoid/togoid-config#396](https://github.com/togoid/togoid-config#396)). We did not

adopt `/search/id` for candidate generation: over 127 probes it returned exactly our local pattern matches, unranked, at one request per identifier.

Preparing LOTUS for RDF Portal

LOTUS (Rutz et al., 2022) curates referenced structure–organism pairs for natural products, but it lives in Wikidata and publishes no RDF of its own. It therefore cannot be added as a database pointing at a live endpoint: the subset is a query pattern rather than a named graph, Wikidata's endpoint stops every query at 60 s (six of LOTUS's twelve published examples exceed that), and the live data carry no version string. We instead wrote a converter, in `scripts/lotus/`, from LOTUS's frozen CSV release (v11, 2026-04-13) to a dated graph that RDF Portal could host. Entities keep their Wikidata IRIs, so the graph joins with IDSM's Wikidata mirror without a mapping table.

The v11 conversion yields 9,138,012 triples covering 672,413 occurrences, 227,256 structures, 37,486 organisms and 91,426 references. Two of the week's lessons came from this work. First, the release's core table is authoritative for which occurrences exist; reading only the richer metadata table silently dropped 48 occurrences. Second, the vocabulary must ship in the same named graph as the data: because TogoMCP queries pin their graph, a vocabulary in a separate graph is invisible, and schema discovery would return nothing. The hand-written vocabulary (4 classes, 44 properties) documents traps in its comments, for example that `lotus:ncbiTaxonId` reaches only 78.0% of organisms, and a test fails if the converter emits a property the vocabulary does not define.

Shuichi Kawashima is curating the converted graph for RDF Portal, which plans to host it after the BioHackathon.

Community, use cases and skills

New MIE authors. Until this week every MIE file had been written by Akira R. Kinjo. Two other people wrote them during BH26. Kozo Nishida contributed four of the eight new databases, working from the [mie-generator skill](#), the MIE specification and the CI checkers; he ran the skill against each endpoint in turn: WikiPathways (<https://sparql.wikipathways.org/sparql>), IDSM (<https://idsm.elixir-czech.cz/sparql/endpoint/idsm>), LIPID MAPS (<https://lipidmaps.org/sparql>) and SwissLipids (<https://beta.sparql.swisslipids.org/sparql>). Yasuhiro Tanizawa, who with Takatomo Fujisawa designed the MarpolBase RDF and set up its SPARQL endpoint, wrote the MarpolBase file, which needed only light revision afterwards, and which is the first MIE written by the maintainer of the database it describes rather than by a consumer of it. Review still caught what tooling could not, such as an example that named the wrong endpoint and would have returned 0 rows if followed literally.

A companion server. Yuki Moriya built TogoCX during the week: an MCP server meant to run alongside TogoMCP rather than replace it. Where TogoMCP hands an agent a schema guide and general SPARQL access, TogoCX returns database edges with reading instructions attached: how the direction of each edge may be stated (as data, measured, assumed or undetermined), what the relation may and must not be used to claim, whether the record states the answer itself, and which queries ran, with their row counts and the reasons the others did not. It targets a failure the MIE files do not address. The problem is rarely that data is missing; it is that the data found does not say what the agent then writes down, as when free text is recorded as a structured field or an answer is used to explain itself. Over four shared cases, adding TogoCX raised gold records recovered from 82% to 95% and cut the share of rows resting on an answer-bearing record from 79% to 64%, while a third measure, competing explanations considered, fell from 74% to 50%, which its author reports as a regression rather than leaving it out. Agents called both servers within the same run, a mean of 7.8 TogoCX calls against 9.2 TogoMCP calls. TogoCX is a research prototype.

Broad-knowledge evaluation. Priscilla Joanne evaluated TogoMCP on BioASQ Task 13b (Tsatsaronis et al., 2015), an expert-curated biomedical question-answering benchmark, to test broad biomedical knowledge alongside the specific data-retrieval use cases for which TogoMCP is designed. Across the 82 yes/no, 95 factoid and 83 list questions, TogoMCP showed no statistically significant decrease in performance relative to the baseline, the same agent equipped with web search but without TogoMCP attached. Further work is needed to turn this into an improvement on broad benchmark questions, but the evaluation supports TogoMCP's value for more specific tasks that require retrieving data from life-science resources.

Ideas from participants. Priscilla Joanne also built and circulated a form asking for databases with SPARQL endpoints to add, general improvements, and use cases where an MCP-based approach makes sense compared with alternatives. It drew no responses. The databases added this week therefore came from the group's own proposals and from conversations at the venue, which is worth recording: at an event where everyone is already deep in their own project, a form competes badly with a hallway conversation.

Skills. At the mid-term report we showed the research-article-analysis skill, which validates a paper's claims about compounds, reactions, pathways and protein functions against ChEBI, Rhea, UniProt, Reactome and GO instead of trusting the paper's text. Such skills used to be installed on each client, so a fix reached nobody who already had a copy, and hosts whose models only call tools could not use them at all. Release 2.18.0 serves the public skills (prism, research-article-analysis and disease-analysis) from the server through two routes that read one directory: a `get_workflow` tool, which lists the workflows, returns a workflow's SKILL.md with its file list and content digest, or returns one of its reference files; and `skill://` resources for hosts that read skills over MCP, in anticipation of the skills extension proposed for MCP (SEP-2640), which the FastMCP versions we use do not yet implement. Because the tool route has no description-based skill triggering, the Usage Guide gained a Workflows section generated from the same registry; this also reaches clients whose cached tool list does not yet show `get_workflow`. Serving skills centrally means that fast-changing facts inside them, of the same kind as MIE gotchas, are corrected for every user at once. Developer-facing skills (mie-generator, qa-generator) are reachable by neither route. The handbook and tutorial now tell readers that the skills come with the connection, so local installation is optional.

Discussion

Table 3 relates the group's objectives to the work reported above.

Table 3: Objectives of the TogoMCP group and corresponding outcomes.

Objective	Outcome
Extend and enhance TogoMCP	11 releases; PubCaseFinder tools; TogoID route suggestions; server-side workflows; log-driven fixes; TogoCX companion server
Examine and refine MIE files	two new MIE authors; result assertions over 420 examples; NANDO, GO, ChEBI, UniProt and LIPID MAPS corrections
Use cases and workflows into skills	research-article-analysis demonstration; three public workflows served by <code>get_workflow</code> (2.18.0)
How to make good SPARQL examples	see below
Add new databases	37 to 45 databases (one an experimental BH26 dataset), 5 on external endpoints and 1 on QLever; LOTUS conversion prepared

Onboarding cost lies in traps, not YAML. Writing an MIE file is quick; finding out where a database returns a plausible wrong answer is not. Every trap in Table 2 would pass a test that only checks that a query runs and returns rows. Quantifying each trap and attaching a re-runnable check is what lets the documentation survive upstream releases, as the NANDO case shows.

A database's own maintainer can write its MIE. The MarpolBase file was written by the person who maintains the database, and needed only light revision in review. That matters more than one file's arithmetic. The expensive part of onboarding is knowing where a database returns a plausible wrong answer, and the people who hold that knowledge are the ones who built the resource; if the format is legible enough for them to write in, the cost of the corpus stops scaling with the time of the team that maintains the server.

Rules learned on one platform do not transfer. Guidance that was true across RDF Portal (pin the graph, federate with SERVICE, beware of literal typing) became endpoint-specific once databases arrived from other operators and other query engines. Agent guidance should state the scope of each rule, and tools that reason about queries, such as the empty-result probe, need the same care. The transfer failure runs in both directions: MarpolBase runs the same engine as RDF Portal and still needed three new rules, one of which (the implicitly regrouped aggregate) turned out to hold on RDF Portal as well and had simply never been provoked. Onboarding a new endpoint is therefore also a test of the documentation already written.

What makes a good SPARQL example. From this week's practice, an example written for an LLM agent should (i) record a machine-assertable result, (ii) name the endpoint it must run on, (iii) avoid the subjects of evaluation questions, (iv) date and check every quantitative claim, and (v) never be copied from an upstream collection without being re-run: SwissLipids' published ID-mapping queries use `rdfs:seeAlso`, which carries only Wikidata links, and return 0 rows, while two of the caveats MarpolBase's own producers ship with the data did not reproduce when we tested them.

Agent-facing APIs should accept what agents already know. Agents reached for E-utilities parameter names and for tool names from cached lists. Accepting aliases and explaining retirements removed failures that no amount of documentation had prevented.

Wrap services only for what the graph cannot compute. The PubCaseFinder tools were justified by ranking and text-mined indexing that SPARQL over the same RDF cannot reproduce; everything else stays a SPARQL query guided by the MIE.

Limitations. We did not re-run the TogoMCP benchmark on the enlarged catalogue during the week, so we cannot yet quantify the effect of the new databases or corrections on answer quality. The checks depend on live external endpoints, which are occasionally unavailable, and all figures are dated measurements that will drift. BH26 Microbes is experimental, so its figures may change with the dataset itself rather than with upstream releases.

Future work

- Add benchmark questions that exercise the eight new databases and re-run the evaluation.
- Host the LOTUS graph on RDF Portal, and report the upstream defects found in LIPID MAPS and other sources to their maintainers.
- Measure use of `get_workflow` from the call logs.
- Promote BH26 Microbes out of experimental status once the dataset's shape is settled.
- Align MIE examples with community SPARQL example collections.
- Onboard more MIE authors, especially among database maintainers, and follow up databases proposed by neighbouring BH26 groups that are building RDF or MCP interfaces.

Software and data availability

- TogoMCP source code: <https://github.com/dbcls/togomcp> (releases v2.12.2 to v2.20.0; MIT License).
- Public TogoMCP server: <https://togomcp.rdfportal.org/>.
- LOTUS converter and vocabulary: `scripts/lotus/` in the TogoMCP repository.
- TogoCX: <https://github.com/moriya-dbcls/togocx-mcp>.
- This report: <https://github.com/arkinjo/BH26-TogoMCP>.

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Author contributions

Akira R. Kinjo: Conceptualization, Software, Data curation, Writing – original draft; Kozo Nishida: Data curation, Software, Validation; Shuichi Kawashima: Data curation, Resources; Yuki Moriya: Software, Validation; Takatomo Fujisawa: Data curation, Software, Resources; Yasuhiro Tanizawa: Data curation, Software, Resources; Priscilla Joanne: Validation, Investigation; Yoko Okabeppu: Data curation, Resources; Shuya Ikeda: Software, Resources; Toyofumi Fujiwara: Software, Resources; Yasunori Yamamoto: Conceptualization, Resources, Writing – review & editing.

References

- Agrawal, A., Balci, H., Hanspers, K., Coort, S. L., Martens, M., Slenter, D. N., Ehrhart, F., Digles, D., et al. (2024). WikiPathways 2024: next generation pathway database. *Nucleic Acids Research*, 52(D1), D679–D689. <https://doi.org/10.1093/nar/gkad960> [cito:citesAsDataSource].
- Aimo, L., Liechti, R., Hyka-Nouspikel, N., Niknejad, A., Gleizes, A., Götz, L., Kuznetsov, D., David, F. P. A., et al. (2015). The SwissLipids knowledgebase for lipid biology. *Bioinformatics*, 31(17), 2860–2866. <https://doi.org/10.1093/bioinformatics/btv285> [cito:citesAsDataSource].
- Aramaki, T., Blanc-Mathieu, R., Endo, H., Ohkubo, K., Kanehisa, M., Goto, S., & Ogata, H. (2020). KofamKOALA: KEGG Ortholog assignment based on profile HMM and adaptive score threshold. *Bioinformatics*, 36(7), 2251–2252. <https://doi.org/10.1093/bioinformatics/btz859> [cito:usesMethodIn].
- Bast, H., & Buchhold, B. (2017). QLever: A Query Engine for Efficient SPARQL+Text Search. *Proceedings of the 2017 ACM Conference on Information and Knowledge Management (CIKM '17)*, 647–656. <https://doi.org/10.1145/3132847.3132921> [cito:citesAsRelated].
- Conroy, M. J., Andrews, R. M., Andrews, S., Cockayne, L., Dennis, E. A., Fahy, E., Gaud, C., Griffiths, W. J., et al. (2024). LIPID MAPS: update to databases and tools for the lipidomics community. *Nucleic Acids Research*, 52(D1), D1677–D1682. <https://doi.org/10.1093/nar/gkad896> [cito:citesAsDataSource].
- Fujiwara, T., Shin, J.-M., & Yamaguchi, A. (2022). Advances in the development of PubCaseFinder, including the new application programming interface and matching algorithm. *Human Mutation*, 43(6), 734–742. <https://doi.org/10.1002/humu.24341> [cito:citesAsAuthority].

- Fujiwara, T., Yamamoto, Y., Kim, J.-D., Buske, O., & Takagi, T. (2018). PubCaseFinder: A Case-Report-Based, Phenotype-Driven Differential-Diagnosis System for Rare Diseases. *American Journal of Human Genetics*, 103(3), 389–399. <https://doi.org/10.1016/j.ajhg.2018.08.003> [cito:citesAsDataSource].
- Galgonek, J., & Vondrášek, J. (2021). IDSM ChemWebRDF: SPARQLing small-molecule datasets. *Journal of Cheminformatics*, 13(1), 38. <https://doi.org/10.1186/s13321-021-00515-1> [cito:citesAsDataSource].
- Galgonek, J., & Vondrášek, J. (2024). The IDSM mass spectrometry extension: searching mass spectra using SPARQL. *Bioinformatics*, 40(4), btac174. <https://doi.org/10.1093/bioinformatics/btac174> [cito:citesAsDataSource].
- Gargano, M. A., Matentzoglou, N., Coleman, B., Addo-Lartey, E. B., Anagnostopoulos, A. V., Anderton, J., Avillach, P., Bagley, A. M., et al. (2024). The Human Phenotype Ontology in 2024: phenotypes around the world. *Nucleic Acids Research*, 52(D1), D1333–D1346. <https://doi.org/10.1093/nar/gkad1005> [cito:citesAsDataSource].
- Ikeda, S., Aoki-Kinoshita, K. F., Chiba, H., Goto, S., Hosoda, M., Kawashima, S., Kim, J.-D., Moriya, Y., et al. (2025). Expanding the concept of ID conversion in TogoID by introducing multi-semantic and label features. *Journal of Biomedical Semantics*, 16(1), 1. <https://doi.org/10.1186/s13326-024-00322-1> [cito:citesAsRelated].
- Ikeda, S., Ono, H., Ohta, T., Chiba, H., Naito, Y., Moriya, Y., Kawashima, S., Yamamoto, Y., et al. (2022). TogoID: an exploratory ID converter to bridge biological datasets. *Bioinformatics*, 38(17), 4194–4199. <https://doi.org/10.1093/bioinformatics/btac491> [cito:usesMethodIn].
- Kinjo, A. R., & Yamamoto, Y. (2026). Measure before you rewrite: ablation-driven redesign of LLM-facing RDF schema documentation in TogoMCP. *BioHackrXiv*. https://doi.org/10.37044/osf.io/6v5ra_v1 [cito:extends].
- Kinjo, A. R., Yamamoto, Y., Bustamante-Larriet, S., Labra-Gayo, J.-E., & Fujisawa, T. (2026). TogoMCP: natural language querying of life-science knowledge graphs via schema-guided LLMs and the Model Context Protocol. *Database*, 2026, baag042. <https://doi.org/10.1093/database/baag042> [cito:citesAsAuthority].
- Kratochvíl, M., Vondrášek, J., & Galgonek, J. (2018). Sachem: a chemical cartridge for high-performance substructure search. *Journal of Cheminformatics*, 10(1), 27. <https://doi.org/10.1186/s13321-018-0282-y> [cito:usesMethodIn].
- Kratochvíl, M., Vondrášek, J., & Galgonek, J. (2019). Interoperable chemical structure search service. *Journal of Cheminformatics*, 11(1), 45. <https://doi.org/10.1186/s13321-019-0367-2> [cito:usesMethodIn].
- O’Leary, N. A., Wright, M. W., Brister, J. R., Ciufu, S., Haddad, D., McVeigh, R., et al. (2016). Reference sequence (RefSeq) database at NCBI: current status, taxonomic expansion, and functional annotation. *Nucleic Acids Research*, 44(D1), D733–D745. <https://doi.org/10.1093/nar/gkv1189> [cito:citesAsDataSource].
- Rutz, A., Sorokina, M., Galgonek, J., Mitchen, D., Willighagen, E., Gaudry, A., Graham, J. G., Stephan, R., et al. (2022). The LOTUS initiative for open knowledge management in natural products research. *eLife*, 11, e70780. <https://doi.org/10.7554/eLife.70780> [cito:citesAsDataSource].
- Tanizawa, Y., Mochizuki, T., Yagura, M., Sakamoto, M., Fujisawa, T., Kawamura, S., Shimokawa, E., Yamaoka, S., Nishihama, R., Bowman, J. L., Berger, F., Yamato, K. T., Kohchi, T., & Nakamura, Y. (2026). MarpolBase: genome database for Marchantia polymorpha featuring high quality reference genome sequences. *Plant and Cell Physiology*, 67(3), 377–388. <https://doi.org/10.1093/pcp/pcf159> [cito:citesAsDataSource].
- Tsatsaronis, G., Balikas, G., Malakasiotis, P., Partalas, I., Zschunke, M., Alvers, M. R., Weissenborn, D., Krithara, A., Petridis, S., Polychronopoulos, D., Almirantis, Y., Pavlopoulos, J., Baskiotis, N., Gallinari, P., Artiéres, T., Ngonga Ngomo, A.-C., Heino, N., Gaussier, E., Barrio-Alvers, L., ... Paliouras, G. (2015). An overview of the BIOASQ large-scale biomedical semantic indexing and question answering competition. *BMC Bioinformatics*, 16, 138. <https://doi.org/10.1186/s12859-015-0564-6> [cito:usesMethodIn].
- Zou, Z., Ohta, T., & Oki, S. (2024). ChIP-Atlas 3.0: a data-mining suite to explore chromosome architecture together with large-scale regulome data. *Nucleic Acids Research*, 52(W1), W45–W53. <https://doi.org/10.1093/nar/gkae358> [cito:citesAsDataSource].